

Naive Bayes: The Smart 'Naive' Choice

Why simplicity beats complexity in machine learning



The Titanic Challenge: Who Survives?

From your notebook: We have data (Class, Sex, Age, Fare). We want to predict 'Survived'. How do we choose the right tool?

How Different Models "Think"



Linear/Logistic

Tries to find a single "line" or boundary that best separates survivors from non-survivors.



K-Nearest Neighbors (KNN)

Asks its "neighbors" for an opinion.
Finds the 'k' most similar passengers
and takes a vote.



Naive Bayes

Plays the "odds" based on evidence.
What's the *probability* of survival
given the passenger's features?

An Interesting Way to Think of Naive Bayes

“

It's like a doctor diagnosing an illness. A fever **probably** means flu. A cough **probably** means flu. Both together? The **probability** of flu is now **much** higher.

”

— Naive Bayes just multiplies these probabilities together.

The "Naive" Assumption (The Secret Sauce)

-  **What is it?** It assumes all features (Age, Class, Sex) are completely **independent** of each other.
-  **In English?** It assumes a passenger's 'Age' has zero effect on their 'Ticket Class'.
-  **Is this true?** Of course not! That's why the algorithm is called "naive".
-  **So why use it?** This one simple, "wrong" assumption makes the math **insanely** fast and surprisingly effective.

Why Naive Bayes vs. Linear/Logistic?

Linear / Logistic Regression

Good at finding complex relationships. Tries to find the *perfect* weighted line. Can be slow to train on massive data.
Can overfit if not careful.

Naive Bayes

Blazing fast training. Doesn't care about relationships, only probabilities. Excellent for *very* wide data (e.g., text with 1000s of words).

Why Naive Bayes vs. KNN?

K-Nearest Neighbors (KNN)

Conceptually simple. No "training" time. *But* prediction is very slow (must compare to *every* point). Fails with high-dimensional data.

Naive Bayes

Instant predictions. Handles high dimensions (many features) with ease. Perfect for real-time systems where speed matters.

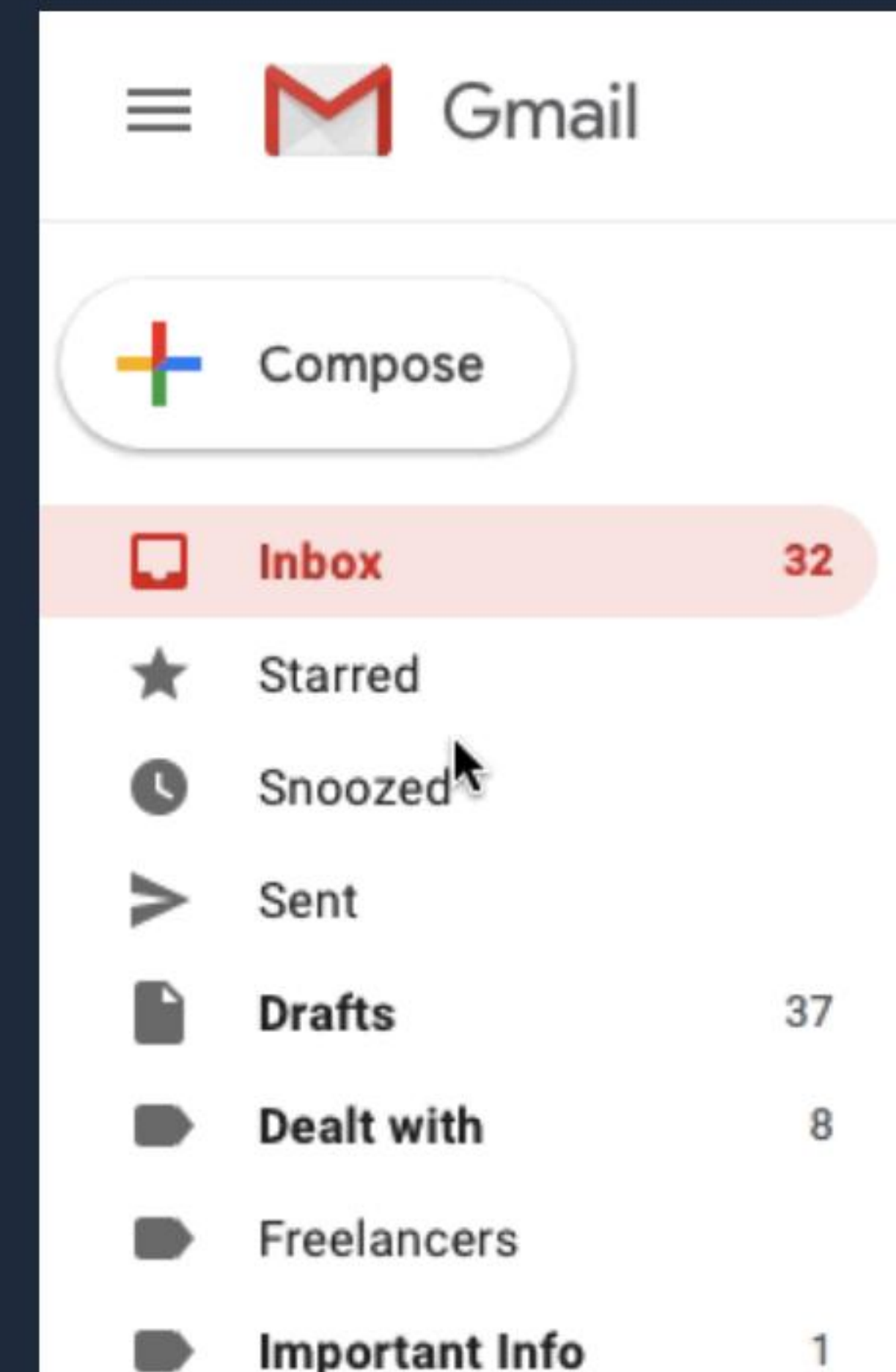


Real-World Expertise

Where Naive Bayes Shines

Real-World Expertise: Spam Filtering

- This is the **classic** Naive Bayes use case.
- It calculates: $P(\text{Spam} \mid \text{word}=\text{'Free'}, \text{word}=\text{'Money'})$.
- The "naive" assumption works well! The word 'Free' is treated as independent of 'Money'.
- It's fast enough to run on every single email you receive in real-time.

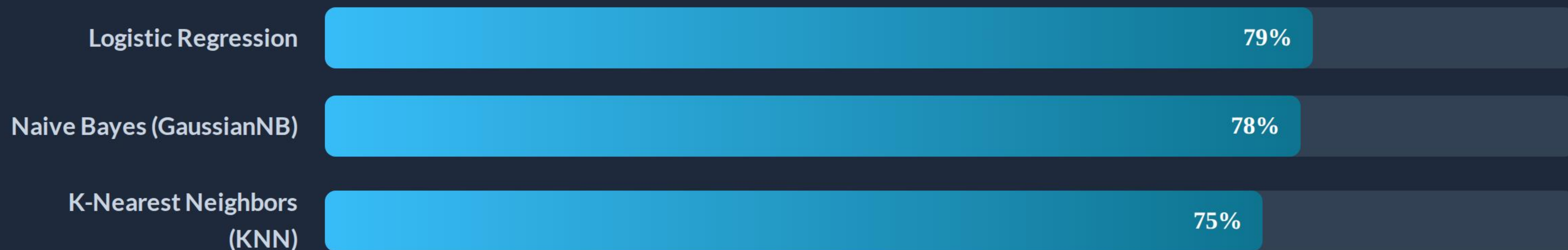


Real-World Expertise: Document Classification

- Is this news article about "Sports" or "Politics"?
- A single article can have thousands of "features" (one for each unique word).
- Naive Bayes excels here, while models like KNN would be too slow and confused by the high dimensions.

digital collage of news article headlines about sports and politics

Titanic Data: Model Performance



*Your notebook score of ~78% is a fantastic *baseline*! It's fast, easy to understand, and gives a great starting point, performing almost as well as the more complex Logistic Regression.*

Questions?

Image Sources



<https://public-assets.typeform.com/helpcenter/uploads/2019/01/21143834/spam-11.gif>

Source: help.typeform.com



<https://blogs.loc.gov/headlinesandheroes/files/2022/12/collage-image-fina-scaled.jpg>

Source: blogs.loc.gov